

Multi-Branch Inventory & Transfer Management System

Entity Relationship Diagram (ERD)

1. Notation Legend

Cardinality uses crow's-foot / line notation. PK rows are shaded gold; FK rows are shaded blue.

	One and only one (mandatory one-to-one)
	One to one-or-many (mandatory one-to-many)
	One to zero-or-many (one-to-many, optional children)
	Zero-or-one to one (optional parent, mandatory child)
PK	Primary Key — uniquely identifies each row
FK	Foreign Key — references the PK of another table
UK	Unique Key — column must be unique across rows

2. Entity Definitions

branches

BRANCHES			
Column	Type	Key	Notes / Constraints
id	BIGINT	PK	Auto-increment surrogate key
name	VARCHAR(100)		Display name of the branch
code	VARCHAR(20)	UK	Short unique identifier e.g. HQ-001
location	VARCHAR(255)		Physical address
contact_info	VARCHAR(255)		Phone / email contact
active	BOOLEAN		FALSE = deactivated (FR-07)
created_at	DATETIME		Record creation timestamp

updated_at	DATETIME		Last modification timestamp
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users

USERS			
Column	Type	Key	Notes / Constraints
id	BIGINT	PK	Auto-increment surrogate key
full_name	VARCHAR(150)		
email	VARCHAR(150)	UK	Used as login credential
password_hash	VARCHAR(255)		bcrypt cost ≥ 12 (FR-04)
role	ENUM		STAFF MANAGER HO_ADMIN ACCOUNTANT ADMIN
branch_id	BIGINT	FK	→ branches.id (NULL for global Admin)
active	BOOLEAN		FALSE = account deactivated (FR-05)
created_at	DATETIME		
updated_at	DATETIME		

items

ITEMS			
Column	Type	Key	Notes / Constraints
id	BIGINT	PK	
name	VARCHAR(150)		Display name
code	VARCHAR(30)	UK	Catalogue code e.g. OFF-001
description	TEXT		Optional long description
category	VARCHAR(100)		Indexed for filtering
unit_of_measure	VARCHAR(50)		e.g. Ream, Box, Unit
unit_price	DECIMAL(15,2)		Used to calculate transfer total_value
active	BOOLEAN		FALSE = deactivated (FR-09)
created_at	DATETIME		
updated_at	DATETIME		

stock_levels

STOCK_LEVELS			
Column	Type	Key	Notes / Constraints
id	BIGINT	PK	
branch_id	BIGINT	FK	→ branches.id (UNIQUE with item_id)

item_id	BIGINT	FK	→ items.id (UNIQUE with branch_id)
quantity_on_hand	INT		Current physical stock at this branch
reserved_quantity	INT		Qty locked by approved in-transit transfers
minimum_threshold	INT		Below this = low-stock alert (FR-12)
last_updated	DATETIME		Auto-updated on any stock change

transfer_requests

TRANSFER_REQUESTS			
Column	Type	Key	Notes / Constraints
id	BIGINT	PK	
source_branch_id	BIGINT	FK	→ branches.id
destination_branch_id	BIGINT	FK	→ branches.id
item_id	BIGINT	FK	→ items.id
quantity	INT		Units requested
total_value	DECIMAL(15, 2)		quantity × unit_price at time of request
justification	TEXT		Required narrative from requester
requires_ho_approval	BOOLEAN		Derived from approval thresholds (FR-15)
status	ENUM		PENDING→MANAGER_APPROVED→HO_APPROVED→IN_TRANSIT→COMPLETED REJECTED CANCELLED
requested_by_id	BIGINT	FK	→ users.id (STAFF or MANAGER)
requested_at	DATETIME		
manager_approved_by_id	BIGINT	FK	→ users.id (MANAGER role)
manager_approved_at	DATETIME		Level-1 approval timestamp
manager_comments	TEXT		Mandatory on rejection
ho_approved_by_id	BIGINT	FK	→ users.id (HO_ADMIN role)
ho_approved_at	DATETIME		Level-2 approval timestamp
ho_comments	TEXT		
dispatched_at	DATETIME		Set when marked IN_TRANSIT
received_at	DATETIME		Set when destination confirms receipt
created_at	DATETIME		
updated_at	DATETIME		

transfer_costs

TRANSFER_COSTS

Column	Type	Key	Notes / Constraints
id	BIGINT	PK	
transfer_request_id	BIGINT	FK	→ <i>transfer_requests.id</i>
amount	DECIMAL(15,2)		
currency	VARCHAR(10)		<i>Default RWF</i>
cost_type	VARCHAR(100)		<i>TRANSPORT HANDLING INSURANCE</i>
notes	TEXT		<i>Free-text description</i>
recorded_by_id	BIGINT	FK	→ <i>users.id (ACCOUNTANT role)</i>
recorded_at	DATETIME		

audit_logs

AUDIT_LOGS			
Column	Type	Key	Notes / Constraints
id	BIGINT	PK	
action	VARCHAR(100)		<i>e.g. TRANSFER_CREATED, MANAGER_APPROVED</i>
entity_type	VARCHAR(100)		<i>e.g. TransferRequest, StockLevel</i>
entity_id	BIGINT		<i>PK of the affected record</i>
performed_by	VARCHAR(150)		<i>Email of acting user (denormalised for immutability)</i>
details	TEXT		<i>Human-readable change description</i>
performed_at	DATETIME		<i>Immutable — no UPDATE/DELETE permitted</i>

system_config

SYSTEM_CONFIG			
Column	Type	Key	Notes / Constraints
id	BIGINT	PK	
config_key	VARCHAR(100)	UK	<i>e.g. APPROVAL_VALUE_THRESHOLD</i>
config_value	VARCHAR(255)		<i>String value; cast to correct type at runtime</i>
description	VARCHAR(500)		<i>Human-readable explanation</i>

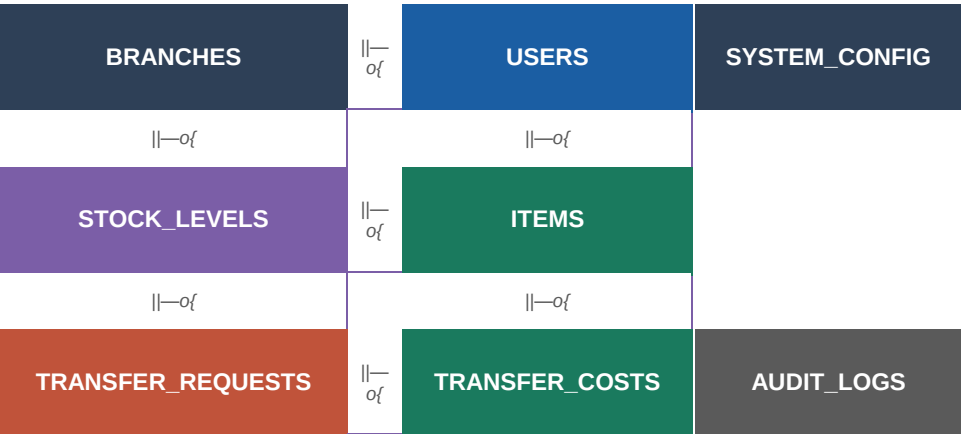
3. Relationships & Cardinality

The table below lists every foreign-key relationship with its cardinality in crow's-foot line notation.

From Entity		Card.	To Entity	Description / FK column
branches	— —o{	users	users.branch_id → branches.id (SET NULL on deactivation)	
branches	— —o{	stock_levels	stock_levels.branch_id → branches.id (CASCADE delete)	
items	— —o{	stock_levels	stock_levels.item_id → items.id (CASCADE delete)	
branches	— —o{	transfer_requests	transfer_requests.source_branch_id → branches.id	
branches	— —o{	transfer_requests	transfer_requests.destination_branch_id → branches.id	
items	— —o{	transfer_requests	transfer_requests.item_id → items.id	
users	— —o{	transfer_requests	transfer_requests.requested_by_id → users.id (requester)	
users	o— —o{	transfer_requests	transfer_requests.manager_approved_by_id → users.id (optional)	
users	o— —o{	transfer_requests	transfer_requests.ho_approved_by_id → users.id (optional)	
transfer_requests	— —o{	transfer_costs	transfer_costs.transfer_request_id → transfer_requests.id	
users	— —o{	transfer_costs	transfer_costs.recorded_by_id → users.id (accountant)	

4. Schema Overview (Relationship Map)

The grid below visualises how all eight tables connect. Arrows are represented by column borders and labels. Read left-to-right and top-to-bottom.



USERS ||—o{ TRANSFER_REQUESTS (requester, mgr_approver, ho_approver) |
TRANSFER_REQUESTS ||—o{ AUDIT_LOGS (via service layer)

Note: USERS connects to TRANSFER_REQUESTS via three FK columns (requested_by_id, manager_approved_by_id, ho_approved_by_id). TRANSFER_REQUESTS also links to BRANCHES twice (source and destination) and to ITEMS once.